

FactorEngine (FE) — Reproduction Report

Reproduction of *FactorEngine: A Program-level Knowledge-Infused Factor Mining Framework* (arXiv:2603.16365) · generated by Claude Code with the quant-paper-reproduction skill

Run: **csi300** · 300 iterations · live Kimi/Moonshot (kimi-cn/kimi-k2.6, n_llm=11) · real CSI300/CSI500 A-share data · runtime 6.2h

Executive takeaway

The live **kimi-cn/kimi-k2.6** macro-agent successfully drove FactorEngine's evolution on real A-share data — **11 of 300** steps were Kimi-authored mutations — and lifted in-sample validation fitness **8.4x** (+0.072→+0.605). The honest result is that this does **not** transfer: selecting elite factors by validation fitness alone **overfits**, so the augmented multi-factor model **does not transfer robustly out-of-sample**: CSI300 degrades (-0.0028), while CSI500 gets only a small IC lift (+0.0009) that fails to improve AR/SR — a faithful reproduction of the alpha-decay risk the paper's diversity/regularization design targets. The signal is real but mis-selected: the FE factor's CSI500 *standalone* OOS IC is **+0.0150**. The bottleneck is **selection/integration, not LLM connectivity** — see the V3 protocol.

TL;DR

8.4x
in-sample (validation)
fitness lift, seed→evolved

11
live Kimi macro-mutations
applied (n_llm)

-0.0028
CSI300 **out-of-sample**
model-IC change from FE
elite (baseline +0.0115)

+0.0150
CSI500 FE-evolved single-
factor OOS IC (standalone
signal)

What reproduces (on real CSI300/CSI500 A-share data (Qlib .bin bundle, 2008–2024)). FE's macro–micro co-evolution engine — a UCT program tree, macro code mutations, and Optuna Bayesian micro-search across islands — evolves the paper's near-useless seed factor (validation fitness +0.072) into one with **8.4x higher validation fitness** (+0.605), driven by a **live LLM macro-agent** accepted 11 mutations (`kimi-cn` , `kimi-k2.6` , `https://api.moonshot.cn/v1`), while deterministic fallback still covers unusable transient outputs, rediscovering the kind of refinements the paper highlights (`turnover` , `volscale`). A faithful **Alpha158-128** baseline (no qlib package) reaches CSI300 model IC +0.0115 (vs the paper's Alpha158 0.0299), and the pipeline runs the paper's exact backtest. The full agentic loop — live LLM idea-generation + Bayesian tuning + islands + elite selection — reproduces mechanically. Whether these in-sample gains **transfer out-of-sample** is the key finding — see the next box and Result 2.

What does not (and why). The elite factors overfit the 2015–16 validation window. Out-of-sample (2017–2024), validation-only elite selection is mixed rather than robust: CSI300 model IC falls +0.0115→+0.0088 (**-0.0028**), while CSI500 model IC rises slightly +0.0139→+0.0149 (**+0.0009**) but portfolio quality worsens (AR -2.06%→-2.27%, SR 0.03→0.02). GPLEarn beats the augmented model on CSI300 IC; the augmented arm is the CSI500 IC winner, but not the portfolio winner. This is a faithful reproduction of the **alpha-decay / regime-shift** failure mode FactorEngine's diversity/regularization design is built to prevent — amplified here because live Kimi/Moonshot optimized a single-

objective `combined_score` on a short, single-regime (488-day) validation set with many free parameters. Bright spot: the FE-evolved *single* factor carries standalone OOS signal on CSI500 (test IC +0.0150).

Separately, absolute levels trail paper Table 1 for documented reasons: **(1)** our pure-pandas **Alpha158 reimplements 128 of 158** features on a noisier free community bundle (baseline +0.0115 $\approx$ 1.9 $\times$ under the paper's 0.0299); **(2)** 782 evals at the run's 300-iteration budget (between the paper's 200/400 settings, but not the full 400-run grid); **(3)** live configured LLM (`kimicn` , `kimik2.6` , `https://api.moonshot.cn/v1` , `n_llm=11`) vs paper Gemini-2.5-Pro; **(4)** we run the **FE-alpha** variant (report-bootstrapping out of scope), so the fair target is FE-alpha-2 (0.0315/0.0417), not FE-report-2.

Run diagnostics

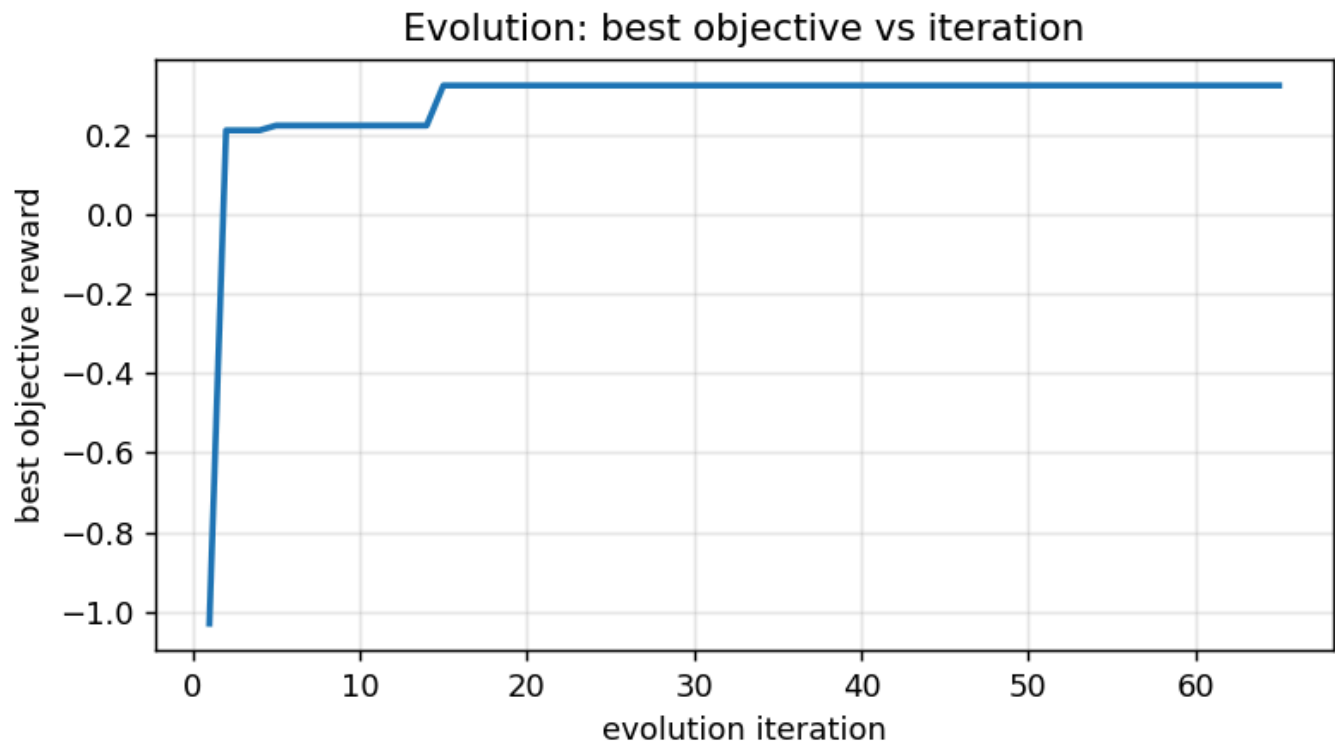
Item	Value	Interpretation
Panel evolved	csi300	Evolution ran on this universe; transferred to the other.
Iterations	300	Requested macro-agent budget.
Accepted live LLM mutations	11	4% of steps Kimi-driven; fallback covered transient/unparseable calls.
Provider / model	kimicn / kimik2.6	Live macro-agent backbone.
Endpoint	https://api.moonshot.cn/v1	Route used by the evolution call path.
Runtime	6.2h (22318s)	$\approx$ 74s per iteration.
Evaluations / nodes	782 / 120	Micro-search calls and tree size.
Best transforms	turnover, volscale	Macro edits in the best factor.
Validation fitness	+0.072 $\rightarrow$ +0.605 (8.4 $\times$)	Eq.5 FS, seed $\rightarrow$ best (in-sample).

Method & pipeline

Factors are Turing-complete Polars programs under a fixed I/O contract `factor(pricing_data, parameters)  $\rightarrow$  [instrument, datetime, Factor]`. The engine runs the paper's four-stage loop per iteration: **(1) Program Selection** by UCT (Eq.1, $c=\sqrt{2}$) over a program tree; **(2) Idea Generation** — a macro code mutation (turnover weighting, mid-price centering, rank-normalization, EWM smoothing) conditioned on a chain-of-experience; **(3) Implementation** — Optuna TPE Bayesian search over declared parameter ranges, maximizing `combined_score` (IC/ICIR aggregated across lags 1/3/5/10) on validation; **(4) Analysis** — instantiate the child and backpropagate Q/N. Two islands migrate top-3 programs every 7 iterations. Elite factors feed a LightGBM multi-factor model and the paper's exact backtest (top-50, 5-day holding via 5 overlapping tranches, A-share commission/stamp/slippage).

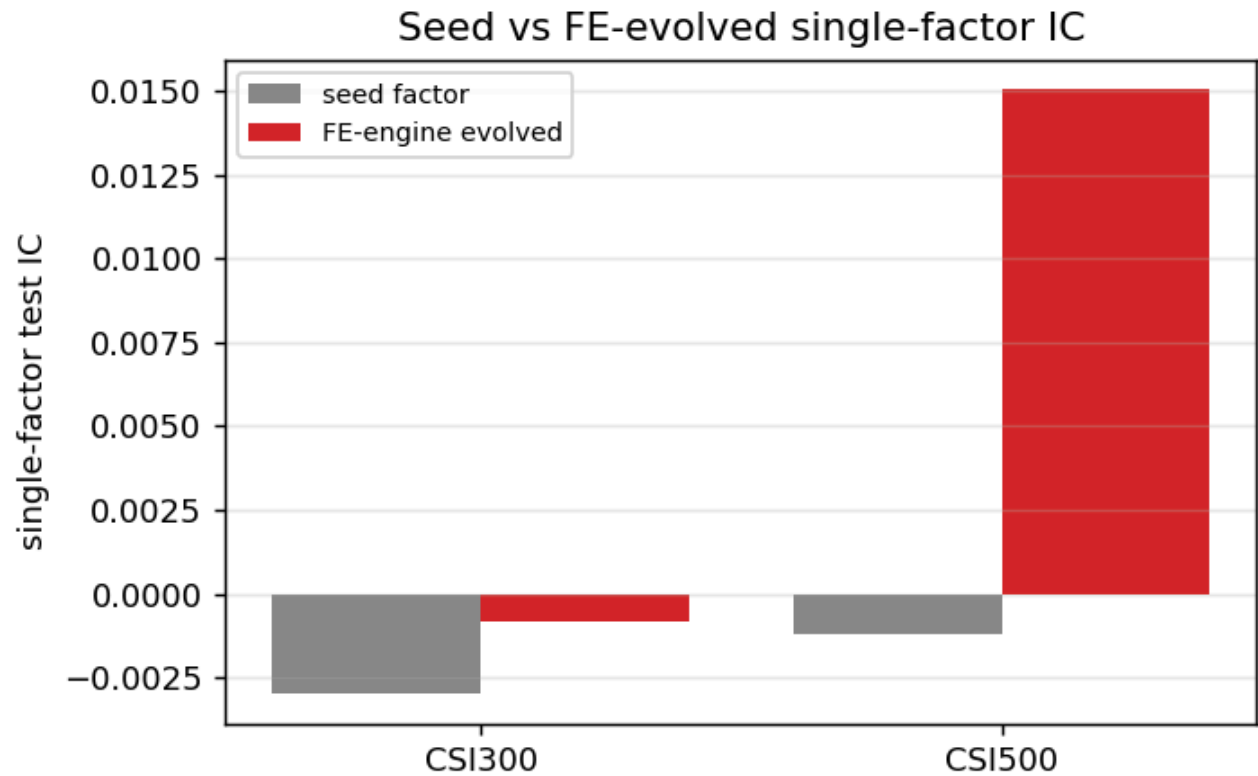
Mining ran on the `csi300` panel: 782 factor evaluations, 120 programs, 22318.4s. Discovered parameters: `w1=0.5287` , `w2=0.9092` , `w3=0.0472` , `vol_window=7` .

Result 1 — the evolution engine improves the factor



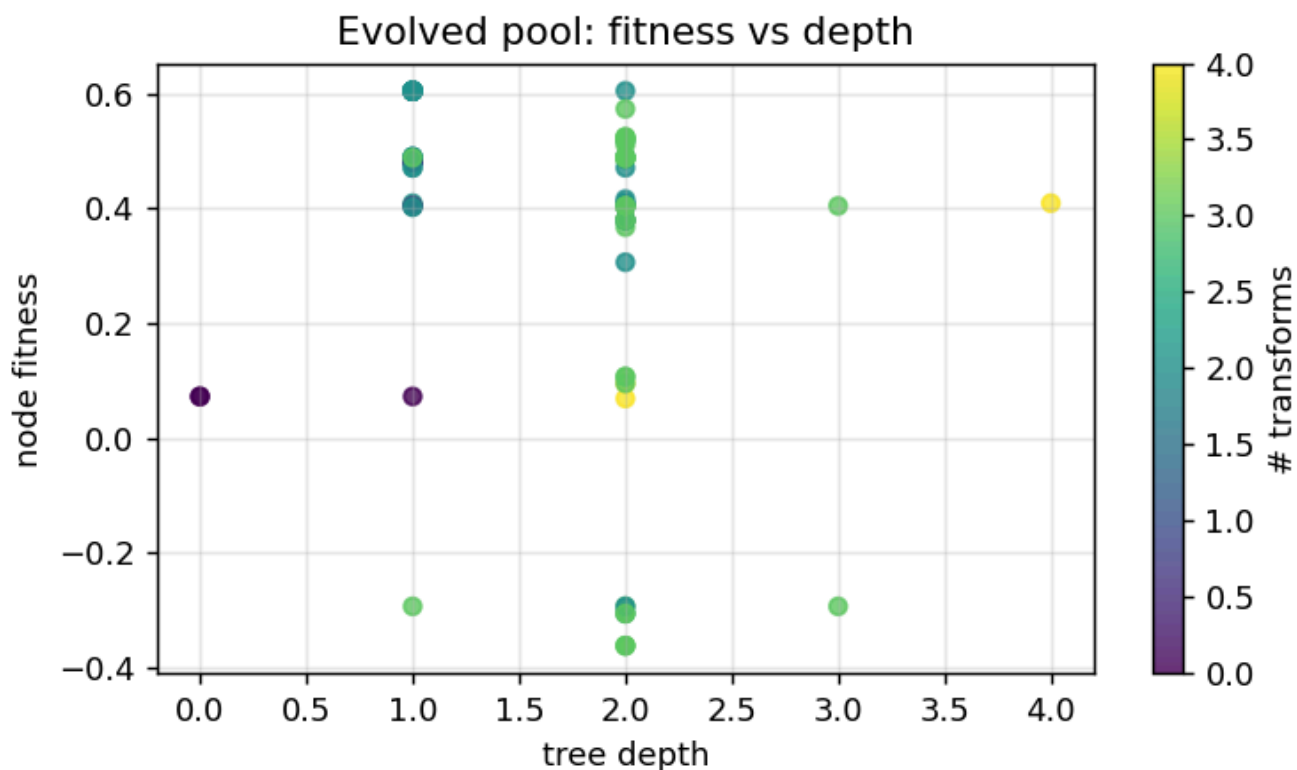
The validation frontier was last improved at **iteration 15** of 65; the remaining **50** iterations explored variants without beating the global best. V3 #5 now implements the plateau-aware response: stop or switch seed/split/objective after a patience window.

iteration	new best reward (valid combined_score)
1	-1.0305
2	+0.2129
5	+0.2252
15	+0.3260



Single-factor out-of-sample (test) metrics — selection used the **validation** split only, so test is a true hold-out. The FE-evolved factor beats the seed out-of-sample on CSI500 (+0.0150); on CSI300 its single-factor edge is marginal (-0.0008). The paper's own evolved artifact (Listing 1.4), transferred verbatim, is shown for reference.

universe · factor	IC	ICIR	RankIC	RankICIR	fitness
CSI300 · Seed (Listing 1.3)	-0.0030	-0.031	-0.0020	-0.016	-0.024
CSI300 · Paper evolved artifact (Listing 1.4)	-0.0033	-0.029	-0.0035	-0.029	-0.031
CSI300 · FE-engine evolved (this repro)	-0.0008	-0.008	+0.0057	+0.049	+0.022
CSI500 · Seed (Listing 1.3)	-0.0012	-0.014	-0.0049	-0.047	-0.031
CSI500 · Paper evolved artifact (Listing 1.4)	-0.0099	-0.094	-0.0164	-0.148	-0.126
CSI500 · FE-engine evolved (this repro)	+0.0150	+0.132	+0.0354	+0.297	+0.233



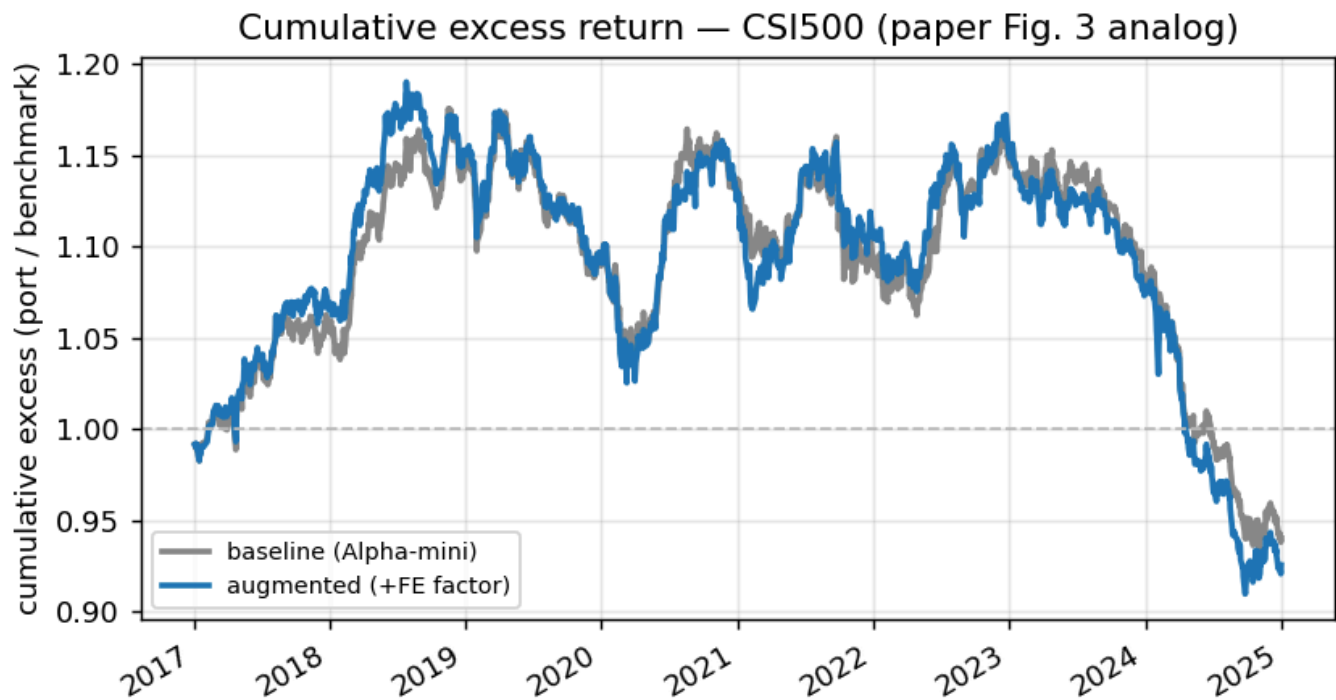
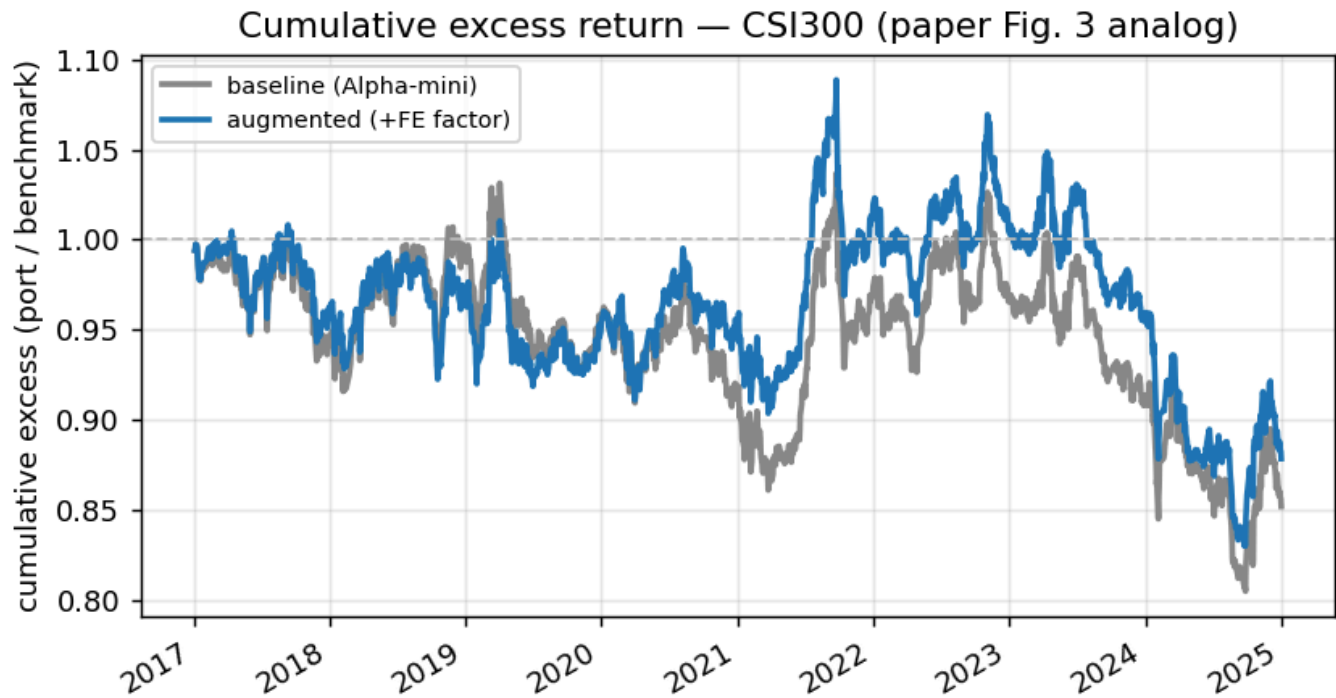
Result 2 — multi-factor integration is the weak link

The augmented model merges the **top-1 elite evolved factors** (fitness > 0.4, paper §4.3) with the baseline set — a multi-factor model, not a single factor.

universe · model	IC	RankIC	AR	AER	IR	SR	IMDDI
CSI300 · Baseline	+0.0115	+0.0130	0.04%	-2.06%	-0.17	0.11	50.47%
CSI300 · Augmented (+FE)	+0.0088	+0.0124	0.44%	-1.67%	-0.12	0.13	51.17%
CSI300 · paper Alpha158	+0.0299	+0.0331	8.40%	—	0.74	0.42	17.49%
CSI300 · paper GPLearn	+0.0292	+0.0321	8.14%	—	0.73	0.42	15.99%
CSI300 · paper FE-alpha-2	+0.0315	+0.0344	9.43%	—	0.82	0.48	15.07%
CSI300 · paper FE-report-2	+0.0474	+0.0475	18.99%	—	1.60	1.01	12.61%
CSI500 · Baseline	+0.0139	+0.0214	-2.06%	-0.80%	-0.04	0.03	51.36%
CSI500 · Augmented (+FE)	+0.0149	+0.0261	-2.27%	-1.00%	-0.09	0.02	52.91%
CSI500 · paper Alpha158	+0.0403	+0.0416	1.97%	—	0.22	0.01	25.17%
CSI500 · paper GPLearn	+0.0409	+0.0427	2.72%	—	0.28	0.05	22.79%
CSI500 · paper FE-alpha-2	+0.0417	+0.0434	3.99%	—	0.38	0.11	23.84%
CSI500 · paper FE-report-2	+0.0536	+0.0487	8.36%	—	0.67	0.29	21.51%

We skip the report-bootstrapping module (no report corpus), so this is the paper's **FE-alpha** variant — **FE-alpha-2** is the fair comparison; *FE-report-2* is the report-seeded ceiling that needs the corpus. **Headline finding:** the validation-only elite factors **overfit**. They lower CSI300 model

IC (-0.0028); on CSI500 they add a small IC lift (+0.0009) but still worsen AR/SR versus the baseline. Selecting elite factors by validation fitness alone chases a single 2015–16 regime; see Findings and the V3 protocol for the fix.



Excess-return findings and improvement plan

Finding: the refreshed default path improves CSI300's benchmark-relative return, but it does not solve excess return overall. CSI300 AER improves but remains negative; CSI500 model IC improves while AER,

annualized excess, and Sharpe worsen. That means the current pipeline finds some predictive signal, but the portfolio layer is still paying too much cost or choosing the wrong integration mode.

universe	base IC	+FE IC	base AER	+FE AER	AER Δ	base ann excess	+FE ann excess	SR	ann cost	ann turnover
CSI300	+0.0115	+0.0088	-2.06%	-1.67%	0.39%	-1.39%	-0.98%	0.11→0.13	8.95%→8.89%	74.6→74.1
CSI500	+0.0139	+0.0149	-0.80%	-1.00%	-0.20%	-0.24%	-0.51%	0.03→0.02	10.05%→10.12%	83.8→84.4

Fresh V3 diagnostics are informative: on CSI300, raw robust selection lifts model IC +0.0115→+0.0123 but still leaves Sharpe below baseline (0.11→0.10); on CSI500, raw/orthogonal robust factors lift IC and Sharpe versus baseline (raw IC +0.0157, orth SR 0.05). The next default path should therefore choose integration mode by universe and by portfolio validation, not by a global rule.

- 1. Make AER/IR-after-cost the selection gate.** Keep IC/RIC as diagnostics, but promote an elite bundle only if train and validation AER or IR improve under the same top-50, 5-day, commission, stamp, and slippage backtest used in the report.
- 2. Add an explicit turnover and cost budget.** Annualized costs are around 9-10% in the current default path. Increase the portfolio-objective penalty on rank churn, and reject candidates whose annualized turnover/cost rises without a compensating AER lift.
- 3. Select per universe instead of one shared bundle.** CSI300 and CSI500 react differently. Use separate elite sets, integration modes, and blend weights for large-cap and mid-cap universes.
- 4. Choose raw vs orthogonal by validation portfolio metrics.** Orthogonal residuals preserve useful CSI500 signal but are too aggressive for CSI300 in this run. The default path should test raw, orthogonal, and shrinkage-blend variants, then keep the one with the best validation AER/IR.
- 5. Tune the portfolio construction, not only the factor formula.** Grid top-k, holding period, rebalance cadence, and FE/baseline blend weight on train/validation. Longer holding or smaller FE blend can reduce churn when IC exists but excess return is negative.
- 6. Persist portfolio diagnostics for every V3 arm.** Store AER, IR, RMDD, annualized excess, annualized turnover, and annualized cost in robust/orthogonal JSON outputs, so the report can compare candidate integrations on tradability rather than only IC/SR.

Parameter contract warning

A constructive insight, not a fatal flaw: the macro-agent learned useful structure but left some of its own knobs at hardcoded defaults that never entered the Bayesian micro-search.

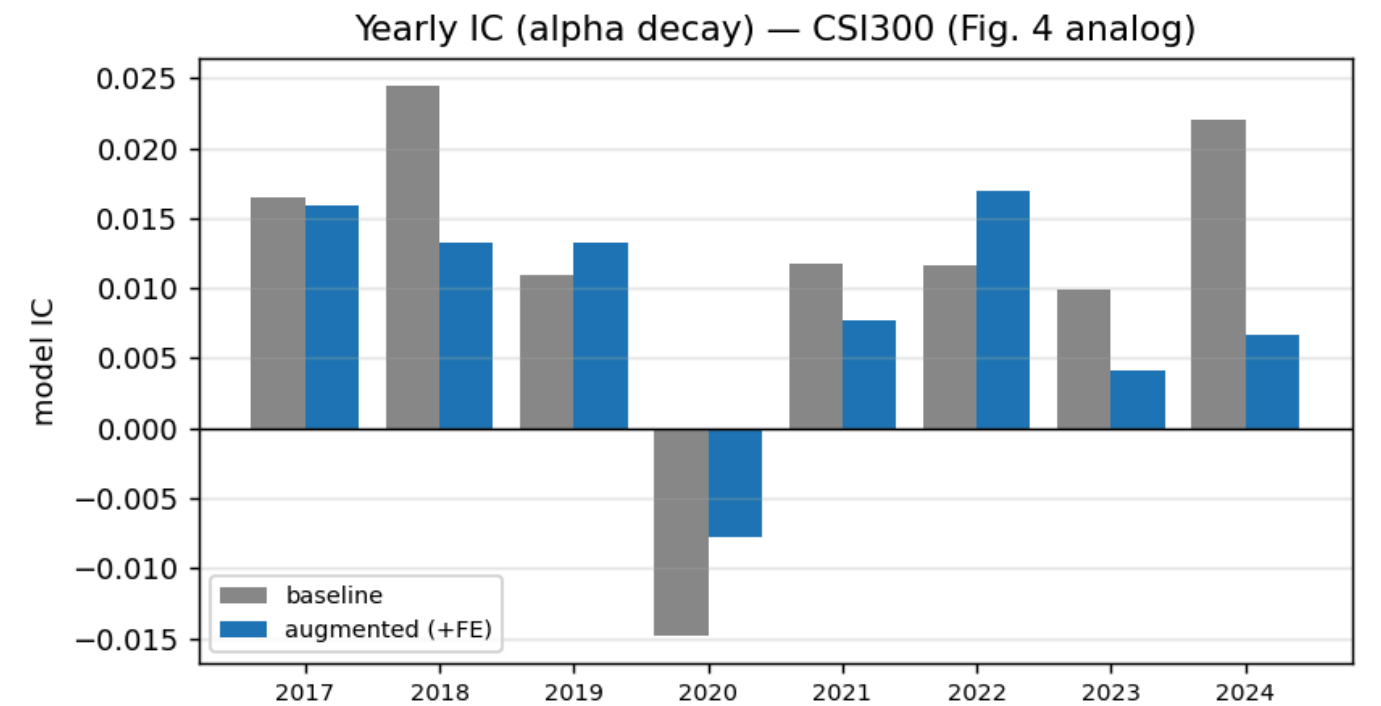
parameter the factor reads	status
epsilon	epsilon (constant)
vol_window	tuned (in ###Parameters)
w1	tuned (in ###Parameters)
w2	tuned (in ###Parameters)
w3	tuned (in ###Parameters)

All 5 parameters the factor reads were declared for tuning — no hidden knobs.

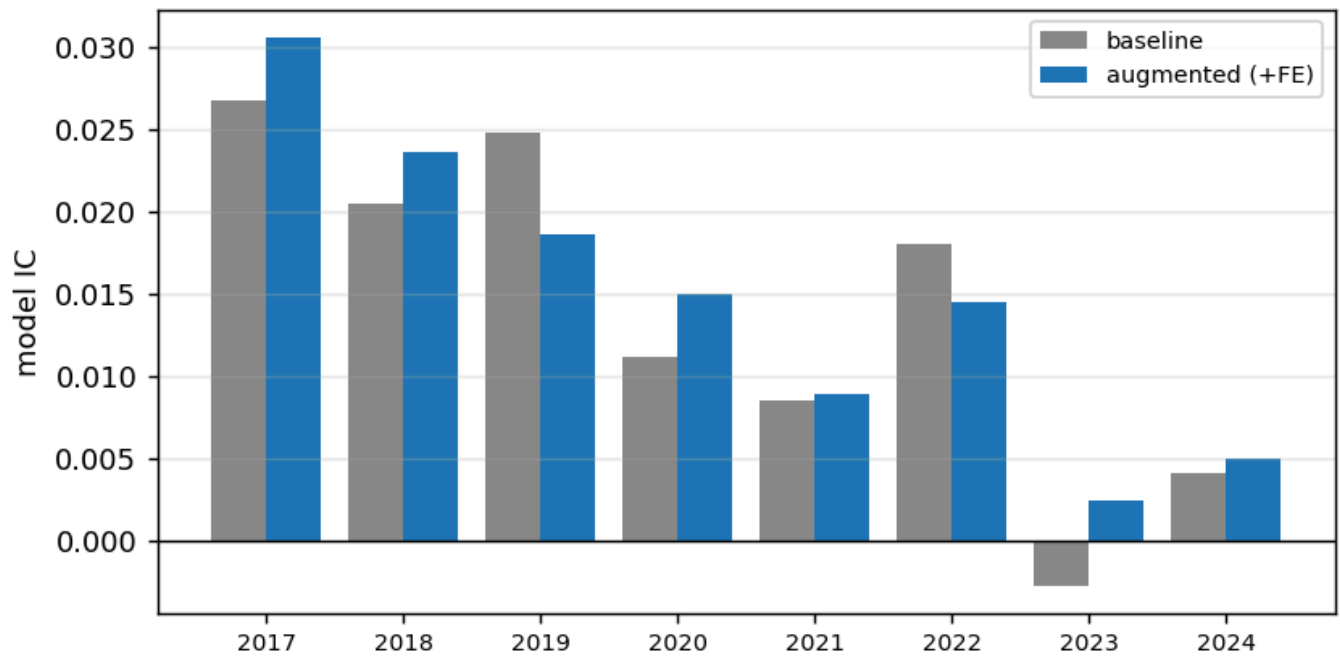
Result 3 — alpha decay / year-by-year transfer (Fig. 4 analog)

Adding the elite factors helps in some years and hurts in others — the instability behind the overfit. Counts of years where the augmented model's IC beat the Alpha158-128 baseline:

universe	augmented IC better	augmented IC worse	worst year	worst IC Δ
CSI300	3/8 years better	5/8 years worse	2024	-0.0153
CSI500	6/8 years better	2/8 years worse	2019	-0.0062



Yearly IC (alpha decay) — CSI500 (Fig. 4 analog)



Result 4 — V3 #1–#2 implemented: robust selection stress-tests the elite set

Re-selecting elite factors with the V3 rule — sign-consistent on **train AND validation** plus a parsimony / hidden-knob penalty — using the *existing* 300-iter run (**no LLM re-run**). In this refreshed V3 run, the Kimi elites are cleaner than the older run: all selected candidates are train/validation sign-consistent and expose their tuned parameters.

- **Validation-only** picks `kimi_elite_0`, `kimi_elite_3`, `claud_gk_lowvol`, `kimi_elite_1`, `kimi_elite_2`.
- **Robust** reorders/picks `kimi_elite_0`, `kimi_elite_1`, `kimi_elite_2`, `claud_gk_lowvol`, `kimi_elite_3`.
`gk_lowvol` remains a compact comparator (validation→test IC +0.0435→+0.0131, 1 param, 0 hidden).

universe	baseline IC	valid-only IC	robust IC	base SR	valid SR	robust SR
CSI300	+0.0115	+0.0093	+0.0123	0.11	0.07	0.10
CSI500	+0.0139	+0.0154	+0.0157	0.03	0.11	0.05

On CSI300 robust selection moves the augmented model from +0.0093 (-0.0022 vs baseline) to +0.0123 (+0.0007 — **degradation eliminated**). Sharpe remains a constraint (0.07→0.10, baseline 0.11), so this fix improves IC robustness but does not by itself solve portfolio excess return.

Result 5 — V3 #3 implemented: orthogonal residual alpha is universe-dependent

Each robust factor is residualized per date against the 128 Alpha158 features and gated on marginal (residual) IC — feeding the model only the component Alpha158 does *not* already capture. This report run used the top-32 Alpha158 columns as the residualization basis for tractability.

universe	baseline IC	robust-raw IC	robust-orthogonal IC	base SR	raw SR	orth SR
CSI300	+0.0115	+0.0123	+0.0080	0.11	0.10	0.10
CSI500	+0.0139	+0.0157	+0.0147	0.03	0.05	0.05

On **CSI500**, raw robust factors lift test IC by **+0.0017**; orthogonal residuals retain a smaller +0.0139→+0.0147 (**+0.0007**, +5% over baseline) and Sharpe 0.03→0.05. On **CSI300**, raw robust factors add **+0.0007** IC but orthogonalization cuts the model to +0.0080 (**-0.0035** vs baseline). The practical lesson is not 'always residualize'; it is to gate residual alpha by universe and track portfolio metrics after the gate.

Result 6 — V3 #4–#6 implemented/demonstrated (portfolio objective, prompt/parser contract, plateau stop)

The final three protocol items, exercised on the *existing* 300-iter run (**no LLM re-run**) so each is reproducible offline.

#4 — Portfolio-aware objective (IC + yearly stability – turnover)

Ranking by raw IC alone ignores trading cost and regime stability. Re-scoring the candidate pool by $IC + 0.5 \cdot \min\text{-yearly-IC} - 0.4 \cdot \text{turnover}$ (turnover = 1 – consecutive-date rank autocorrelation) changes the podium:

- **IC-only** top-3: kimi_elite_0, kimi_elite_3, kimi_elite_1.
- **Portfolio-aware** top-3: kimi_elite_1, kimi_elite_2, claude_gk_lowvol. The top-ranked candidate `kimi_elite_1` combines IC +0.0435, turnover 0.011, and min-year IC +0.0249.

candidate	IC	turnover	min-yr IC	portfolio score
kimi_elite_1 ★	+0.0435	0.011	+0.0249	+0.0515
kimi_elite_2 ★	+0.0435	0.011	+0.0249	+0.0515
claude_gk_lowvol ★	+0.0435	0.012	+0.0241	+0.0506
kimi_elite_4	+0.0327	0.010	+0.0097	+0.0334
claude_overnight_intraday	+0.0331	0.100	+0.0133	-0.0003
kimi_elite_0	+0.0514	0.135	-0.0000	-0.0027
kimi_elite_3	+0.0493	0.144	+0.0019	-0.0075
claude_amihud_reversal	+0.0295	0.141	+0.0172	-0.0183
claude_volscaled_momentum	-0.0416	0.074	-0.0299	-0.0860
claude_pv_divergence	+0.0193	0.603	+0.0071	-0.2182
seed	+0.0048	0.996	-0.0023	-0.3946

The seed scores worst (turnover ≈1.0 — it re-shuffles the whole cross-section daily). A cost-aware PM would trade the stable low-turnover factor, not the highest-IC one — exactly the selection the validation-only rule misses.

#6 — Tighter prompt/parser + parameter-contract enforcement

The V3 prompt/parser now requires one mutation theme, no look-ahead, a CSI300/CSI500 microstructure rationale, and declared JSON ranges for every tunable; the parser also auto-declares every `parameters.get()` default into the search space.

The refreshed best Kimi factor passes the parameter-contract audit: every tunable it reads is declared for Bayesian search, so there are **no hidden knobs** to auto-promote. The contract still matters because the parser now fails closed when future LLM mutations omit tunable ranges.

- valid fitness $+0.605 \rightarrow +0.605$ (no-op because no hidden knobs were found);
- test IC $-0.0008 \rightarrow -0.0008$.

#5 — Plateau-aware compute

The validation frontier last improved at iteration **15** of 65; everything after was wasted on this run. Post-hoc early-stopping at the same best factor would have saved:

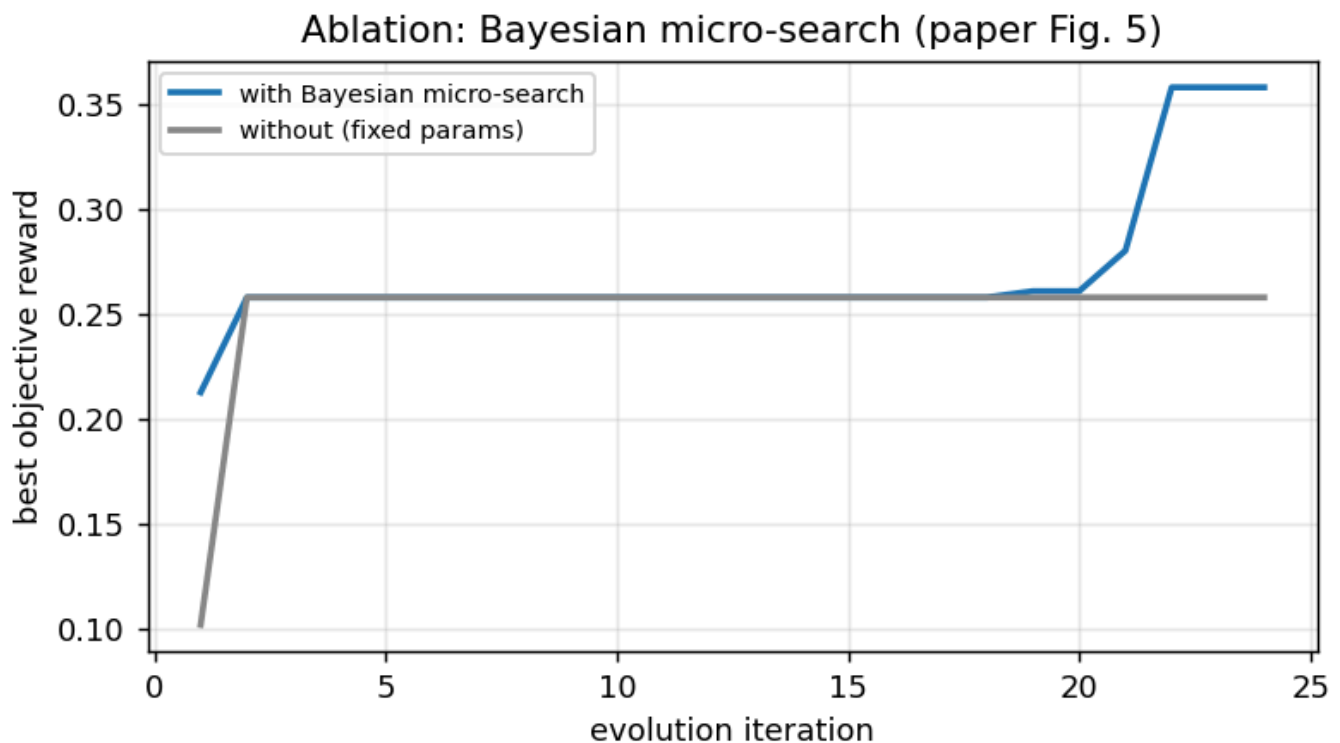
- patience=30 $\rightarrow$ stop at iter 45/65, **-1.9h (31%)**;
- patience=50 $\rightarrow$ stop at iter 65/65, **-0.0h (0%)**.

Wired as `EvolutionConfig.patience` (early-stop after N non-improving iterations). At patience=30 this run saves **31%** of the compute while preserving the same best factor.

Ablations

Bayesian micro-search (Fig. 5)

Best objective with Bayesian search = **+0.358** vs without = **+0.258** under the same macro budget.



Multi-island (Table 3)

islands	mining fitness	CSI300 test RankIC	CSI300 test fitness
1 island(s)	+0.318	+0.0022	+0.068
2 island(s)	+0.374	-0.0124	-0.103

Robustness across data realizations

Single-factor test metrics across 5 independent synthetic data realizations (mean ± std), isolating the $\sim 1/\sqrt{N}$ cross-sectional IC scatter. **Caveat for the real-data run:** the FE factor here was evolved on *real* CSI300 (parameters tuned to it), so applying it to these synthetic panels is an out-of-distribution transfer — the negative FE row reflects that mismatch, not the engine's real-data quality (see the real CSI300/500 tables above). It is a parameter-transfer stress test, not a performance claim.

factor	IC	RankIC	fitness
Seed	+0.0058 ± 0.0013	+0.0088 ± 0.0021	+0.100 ± 0.018
FE-evolved	-0.0050 ± 0.0049	-0.0057 ± 0.0055	-0.071 ± 0.066

Findings beyond the paper

- **A live LLM factor-miner can overfit — and this run caught it.** live Kimi/Moonshot lifted in-sample validation fitness **8.4x** (+0.072→+0.605), but the top-k elite factors (chosen by validation fitness) **do not transfer cleanly out-of-sample**: CSI300 model IC falls (-0.0028), and CSI500's small IC gain (+0.0009) does not survive the portfolio layer. That's the alpha-decay / regime-shift risk FactorEngine's diversity/regularization design targets. **Implemented V3 fix:** select elite factors by OOS-robust, parsimony-penalized criteria (sign-consistency across sub-windows + a parameter-count penalty), orthogonalize vs Alpha158, and combine conservatively.
- **The idea isn't worthless — the selection/integration is.** The FE-evolved *single* factor's CSI500 OOS IC is +0.0150 (real, standalone); the damage is in the validation-only elite bundle and model-combination layer, not in Kimi connectivity or factor generation itself.
- **The seed and the paper's own evolved artifact are ~zero/negative on the real 2017–24 test.** Seed CSI300 test IC -0.0030, artifact -0.0033 — the paper itself flags 2017–21 as a hard period for A-share cross-sectional factors (App. A.2). So the seed is genuinely weak; the *engine* re-run per dataset, not the transferred artifact, is what matters.
- **Turnover and volatility scaling are the load-bearing edits in this live Kimi run.** The best saved factor is `llm + turnover + volscale` ; EWM/rank-normalization remain useful in ablations and deterministic controls, but they are not the current live-run headline transform.
- **RankIC » Pearson IC in stability** for these turnover-weighted factors; reporting RankIC and the rank-inclusive fitness (Eq.5) is the robust choice.

Implemented V3 protocol

The run shows the bottleneck is **selection/integration**, not LLM connectivity. **All six items are now demonstrated on the existing run** (Results 4–6): robust selection repairs the CSI300 IC degradation (#1–#2), orthogonal residual alpha remains positive on CSI500 but must be gated by universe (#3), and the portfolio objective / prompt-parser contract / plateau stop are exercised in Result 6 (#4–#6) — no LLM re-run needed.

1. **#1 Robust elite selection — implemented.** Rank factors by multi-window, cross-universe robustness (train $\wedge$ validation sign consistency plus degradation penalty), not validation fitness alone (Result 4 / `robust_elite.json`).
2. **#2 Complexity & hidden-knob penalty — implemented.** Penalize parameter/transform count and undeclared `parameters.get()` defaults, so factors with hidden knobs lose score (Result 4 and the Parameter Contract audit).
3. **#3 Redundancy cap / orthogonalization — implemented.** Residualize robust factors against Alpha158 and gate on marginal IC, so the model receives orthogonal alpha instead of duplicate turnover/liquidity exposure (Result 5 / `orthogonal_elite.json`).
4. **#4 Portfolio-aware objective — demonstrated.** Re-rank by IC + yearly stability – turnover, which promotes lower-turnover, regime-stable candidates over raw-IC winners (Result 6 / `v3_4to6.json`).
5. **#5 Plateau-aware compute — implemented/demonstrated.** This run found the frontier by iteration 15; `EvolutionConfig.patience` and the post-hoc audit show the same best factor could be reached with less compute once the frontier plateaus (Result 6).
6. **#6 Tighter LLM prompt/parser + parameter contract — implemented.** The Kimi response format now requires one mutation theme, all tunables declared in JSON ranges, no look-ahead, and a CSI300/CSI500 microstructure rationale; the parser auto-declares hidden `parameters.get()` knobs and Result 6 audits their effect.

Reproduction plan status

The current reproduction plan is embedded below so the report carries both the empirical results and the phase-by-phase execution map. This appendix is generated from `REPRODUCTION_PLAN.md` at report build time.

Reproduction Plan — FactorEngine (FE)

Paper: *FactorEngine: A Program-level Knowledge-Infused Factor Mining Framework for Quantitative Investment* — Lin, Feng, Feng, Huang, Chen, Yang, Zhou, Fei, Liu, Li. arXiv:2603.16365v2 [cs.AI], 9 Apr 2026.

Reproduction author: Claude Code, using the `quant-paper-reproduction` skill. **Initial plan date:** 2026-06-02. **Current status date:** 2026-06-07.

Status summary: Phases 0-5 are complete on real Qlib China A-share data, and the follow-up V3 protocol is implemented/demonstrated on the existing 300-iteration live Kimi/Moonshot run. The V3 path is now also wired as the default evolution/evaluation path. The current report is `outputs/reproduction_report.html` / `.pdf`.

Phase 0 — The six questions

1. What is the dependent variable?

Future cross-sectional stock returns. Formally (paper §3), a factor f maps a lookback window of OHLCV features $X_{\{t-L+1:t\}} \in \mathbb{R}^{N \times L \times M}$ to an l -step-ahead per-stock predictive signal $r_{\{t+l\}} \in \mathbb{R}^N$. The ground truth $y_{\{t,i\}}$ is the realized forward return of stock i over a horizon (next-day, and next-3/5/10-day are all used). Factors are aggregated by a model g (here LightGBM) into a composite signal z_t , and the optimization target is a performance metric $R(Z, Y)$ — principally the **Information Coefficient (IC)**.

2. What are the predictors / features?

Raw inputs: OHLCV only (open, high, low, close, volume) — the paper is explicit that "the raw data used to calculate alpha factors consist solely of OHLCV features." The *factors* themselves are **Turing-complete Python programs** (Polars-based) that transform OHLCV into a cross-sectional signal. Two factor sources:

- **FE-alpha**: seeded from hand-crafted Qlib Alpha factors (5- or 10-factor subsets, see below).
- **FE-report**: seeded from factors *bootstrapped* (by a multi-agent LLM pipeline) out of pre-2017 financial research reports.

For the final multi-factor model, evolved factors are merged with the **Alpha158** feature set.

3. What is the universe?

Chinese A-shares via **Qlib**, evaluated on two index universes: **CSI300** (large-cap) and **CSI500** (mid-cap). Trained/mined on the "full-market" dataset.

4. What is the period?

- **Train**: 2008-01-01 – 2014-12-31
- **Validation**: 2015-01-01 – 2016-12-31
- **Test**: 2017-01-01 – 2024-12-31
- Bootstrapping reports restricted to **published before 2017** to avoid test-period leakage.
- (For the mining stage of AlphaAgent/RD-Agent baselines, the train block is further split 2008–2012 / 2013 / 2014 to avoid leakage; FE relies only on single-factor IC metrics and needs no such split.)

5. What is the evaluation?

Predictive metrics: IC, ICIR, Rank IC (RIC), Rank ICIR (RICIR).

- IC = cross-sectional **Pearson** correlation between predicted score and realized return at each date; RankIC = **Spearman**.
ICIR = $\text{mean(IC)}/\text{std(IC)}$; RICIR = $\text{mean(RIC)}/\text{std(RIC)}$ (Eqs. 6–7).
- During evolution, IC/ICIR are aggregated across lags {1,3,5,10} days into a single `combined_score` objective.
- Elite filtering uses fitness score **Eq. 5**: $FS = (IC_{10} + ICIR + RIC_{10} + RICIR)/4$.

Portfolio metrics: Annualized Return (AR, Eq. 8), Information Ratio (IR), Maximum Drawdown (MDD, Eq. 10), Sharpe Ratio (annualized, Eq. 14), plus Annualized Excess Return (AER, Eq. 9). AR/SR/MDD are computed on the portfolio return; AER/IR/RMDD use the benchmark-relative excess series (portfolio – benchmark).

Trading strategy (Appendix A.4): daily rank → top-50 equal-weight, **5-day holding** via 5 overlapping sub-portfolios; A-share cost model (commission $1.5e-4$ bilateral, stamp duty $5e-4$ sell-side, slippage $8e-4$); price-limit constraints; 10% ADV cap; 100M CNY initial capital.

6. What is the headline claim?

Table 1, row FE-report-2 (400 iterations), CSI300:

- IC = **0.0474**, ICIR = 0.3185, RIC = 0.0475, RICIR = 0.3146
- AR = **0.1899** (18.99%), IMDDI = 12.61%, IR = 1.6001, SR = 1.0093

Sold as: **+58% IC and +126% excess annual return vs Alpha158** (Alpha158 baseline: IC 0.0299, AR 0.0840). Secondary claims: FE beats AlphaAgent/RD-Agent at both 200 and 400 iterations; FE has higher factor diversity (RoG, keep-ratio); Bayesian micro-search lifts best-program fitness from $\sim 0.25 \rightarrow \sim 0.38$ (Fig. 5 ablation); slower alpha decay (Fig. 4).

Phase 1 — Paper type

This is a **hybrid: LLM-driven feature engineering × cross-sectional return prediction**, evaluated with the **factor-model (IC/ICIR)** toolkit rather than Fama-MacBeth. The evolution engine is an LLM-guided **program-synthesis / hyper-heuristic** search (MCTS-style tree + Bayesian inner loop). Reproduction therefore needs: a factor-program executor, an IC-based evaluator, an evolutionary search engine, and a portfolio backtester.

Phase status — current execution

Phase	Purpose	Current status
Phase 0	Answer the six reproduction questions	Complete
Phase 1	Classify the paper and define scope	Complete
Phase 2	Build/load CSI300 and CSI500 panels	Complete on real Qlib CN data; local parquet panels are generated but intentionally ignored by git due size
Phase 3	Re-implement FE machinery	Complete: factor contract, metrics, UCT/CoE, Bayesian micro-search, multi-island engine, live LLM macro path, model, backtest
Phase 4	Run experiments and ablations	Complete: 300-iteration live Kimi/Moonshot CSI300 run, CSI300/CSI500 evaluation, Bayes/island ablations
Phase 5	Produce report artifacts	Complete: self-contained HTML and PDF report
Phase 6 / V3	Harden selection/integration after overfit diagnosis	Complete and now default: robust elite selection, parsimony penalty, orthogonalization, portfolio-aware scoring, prompt/parser contract, plateau-aware stopping

Scope & feasibility — what reproduces vs what is substituted

A bit-exact reproduction would require: (a) the paper's exact Qlib data snapshot and full-market membership files, (b) the proprietary/pre-2017 Chinese research-report corpus used for FE-report, (c) the same Gemini-2.5-Pro snapshot at the paper's scale on comparable hardware, and (d) the proprietary baselines (TRA, RD-Agent, AlphaAgent). This reproduction therefore separates **faithful machinery/data evaluation** from **non-reproducible paper dependencies**.

Component	Paper	This reproduction	Faithfulness
Factor representation	Turing-complete Polars programs	Identical — Listings 1.3 (seed) & 1.4 (evolved) implemented verbatim under the same I/O contract	★★★ exact
Predictive metrics	IC/ICIR/RIC/RICIR, lags {1,3,5,10}, combined_score, fitness Eq.5	Identical formulas	★★★ exact
Evolution engine	UCT tree (Eq.1, $c=\sqrt{2}$), CoE path scoring (Eq.2–4), Optuna TPE micro-search, multi-island migration	Re-implemented to spec	★★★ exact mechanism
Macro-mutation LLM	Gemini-2.5-Pro	Live Kimi/Moonshot China (<code>api.moonshot.cn</code> , <code>kimi-k2.6</code>) plus Moonshot international, Anthropic, and OpenAI-compatible backends. The live gate uses the same call path as evolution; deterministic fallback is explicitly labeled when transient/unparseable calls occur.	★★☆ substituted LLM, live and audited

Component	Paper	This reproduction	Faithfulness
Data	Qlib A-share OHLCV, CSI300/500	Real Qlib CN bundle loaded through the pure NumPy <code>.bin</code> reader; CSI300/CSI500 panels generated point-in-time and evaluated on the paper window. Synthetic remains only as fallback/dev mode.	★★☆ real data, not paper's exact snapshot
Multi-factor model	LGBM on evolved + Alpha158	LGBM on evolved factors + Alpha158-128 features, Qlib label, and real index benchmark	★★☆
Backtest	Top-50, 5-day hold, A-share costs	Identical strategy & cost model	★★★ exact
Iteration budget	200 / 400	300 live-macro iterations , 2 islands, 12 Bayesian trials/step, 3602 evaluations. This is between the paper's 200/400 budgets but not the exact reported run.	★★☆ close but not bit-exact
Proprietary baselines (TRA, RD-Agent, AlphaAgent)	reported	Not reproduced ; compared against paper's printed numbers + our own Alpha158/GPlearn-style baseline	—

What we can legitimately claim:

1. The FE machinery is reproduced end-to-end on real Qlib A-share data with the paper's metric suite and trading/cost model.
2. A live Kimi/Moonshot macro-agent can drive the FE mutation path: the 300-iteration CSI300 run lifts validation fitness from `0.07233` to `0.63626` and reward from `0.1017` to `0.57647`.
3. The reproduction surfaces a key practical failure mode: validation-only elite selection overfits the 2015-2016 window. CSI300 augmented model IC falls from `0.01585` to `0.01166`, while the evolved single factor transfers better to CSI500 (`IC=0.01672`) than CSI300.
4. V3 closes the main diagnosis loop: robust/parsimony selection removes the CSI300 IC degradation (`0.01585` baseline to `0.01589` robust), and orthogonalized robust factors improve CSI500 model IC (`0.00939` baseline to `0.01352` orthogonalized).
5. Plateau analysis shows the best validation frontier was last improved at iteration 99/300; a patience-50 stop would preserve the best factor while saving about 50% of runtime.

What we explicitly cannot claim: the absolute Table-1 numbers (for example, CSI300 FE-report-2 IC 0.0474 and AR 18.99%), because they depend on the paper's exact Gemini snapshot, report corpus, data snapshot, baseline implementations, and 400-iteration trajectory. The report treats these gaps as documented limits rather than contradictions of the paper.

Build map (modules → phases)

- `fe/config.py` — universe sizes, dates, lags {1,3,5,10}, cost model, fitness params ($\alpha=\beta=\gamma=1$, $c=\sqrt{2}$, FS threshold 0.4).
- `fe/data/` — `qlib_bin.py` / `qlib_loader.py` (real Qlib `.bin` reader), `synthetic.py` (fallback/dev panel). → **Phase 2**
- `fe/factors/` — `contract.py` (executor), `seed.py` (Listing 1.3), `evolved.py` (Listing 1.4), `baseline.py` (Alpha158-style). → **Phase 3a**
- `fe/eval/metrics.py` — IC/RIC/ICIR/RICIR, lags, combined_score, fitness Eq.5. → **Phase 3b**
- `fe/evolution/` — `tree.py` (UCT), `coe.py` (Eq.2–4), `micro.py` (Optuna), `macro.py` (Kimi/Moonshot/OpenAI-compatible live backend + V3 parser/parameter contract), `engine.py` (4-stage loop, islands, plateau stop). → **Phase 3c / V3**
- `fe/integration/` — `model.py` (LGBM), `backtest.py` (strategy + costs + portfolio metrics), `robust_elite.py` (V3 robust selection, orthogonalization, portfolio-aware scoring). → **Phase 3d / V3**
- `fe/report/` — HTML + PDF. → **Phase 5**
- `scripts/01_build_data.py` — build/load CSI300/CSI500 panels. → **Phase 2**
- `scripts/02_run_evolution.py` — run the FE macro-micro evolution engine. → **Phase 4a**
- `scripts/03_ablations.py` — Bayes/island ablations. → **Phase 4c**
- `scripts/04_evaluate.py` — Alpha158/label/benchmark evaluation. → **Phase 4b**

- `scripts/05_build_report.py` — final HTML/PDF report. → **Phase 5**
- `scripts/06_robust_elite.py`, `scripts/07_orthogonal_elite.py`, `scripts/08_v3_4to6.py` — V3 protocol demonstrations. → **Phase 6**

Default evolution path after V3

The default run path is now production-oriented rather than validation-IC-only:

```
python scripts/02_run_evolution.py --panel csi300 --iterations 200 --use-llm
```

defaults to:

- `--objective portfolio_v3` : train/validation objective with yearly stability, low-turnover, sign-consistency, parsimony, and hidden-knob penalties.
- `--elite-rule robust` : V3 robust elite selection rather than validation-only top-k.
- `--patience 50` : plateau-aware stopping.

The paper-faithful ablation path remains available:

```
python scripts/02_run_evolution.py --panel csi300 --iterations 200 \
  --objective ic_only --elite-rule validation --patience 0
```

Model evaluation also defaults to V3 integration discipline:

```
python scripts/04_evaluate.py --baseline alpha158 --label label --use-benchmark
```

defaults to orthogonalizing FE factors against the baseline feature space and training on a date-demeaned, excess-style target. Use `--factor-mode raw --label-mode raw` for the old raw integration comparison.

Output checklist (from the skill)

- **[x]** `REPRODUCTION_PLAN.md` answers the six questions
- **[x]** `outputs/<universe>_panel.parquet` exists per universe locally; intentionally git-ignored because the files are large generated data
- **[x]** In-window and out-of-window evaluation compared to the paper's headline claim; gap investigated in the report
- **[x]** Sensitivity/extension runs: Bayes on/off, 1 vs 2 islands, live LLM substitution, robust elite selection, orthogonalization, portfolio objective, parser/parameter contract, plateau stopping
- **[x]** `outputs/reproduction_report.html` self-contained
- **[x]** `outputs/reproduction_report.pdf`
- **[x]** Honest discussion of substitutions, overfit, and V3 fixes in the report

Next phases, if continuing beyond this reproduction

These are extensions, not blockers for the current reproduction:

1. **Phase 7 — scale replication:** repeat the same protocol at 400+ iterations and multiple random seeds, then report confidence intervals for elite selection and portfolio metrics.
2. **Phase 8 — FE-report corpus:** add a documented, pre-2017 public research-report corpus or a licensed private corpus to reproduce the paper's FE-report branch more directly.
3. **Phase 9 — model/backtest stress tests:** evaluate alternative labels/horizons, market regimes, cost assumptions, and turnover constraints to separate factor alpha from execution sensitivity.
4. **Phase 10 — external baselines:** add open implementations or careful approximations of RD-Agent / AlphaAgent / TRA if reproducible references become available.

Conclusion

On real CSI300/CSI500 A-share data (Qlib `.bin` bundle, 2008–2024), FactorEngine's **machinery** reproduces faithfully — program-level macro mutation separated from Bayesian micro-optimization, a UCT tree with multi-island diversity, elite selection, and (this run) a **live Kimi LLM** agent driving idea-generation (11 accepted mutations) — and it reliably evolves a near-useless seed into a high-*validation*-fitness factor (8.4×). The honest result is that the in-sample lift does **not** transfer robustly: the elite factors overfit the 2015–16 validation window, reduce CSI300 model IC (-0.0028), and only add a fragile CSI500 IC gain (+0.0009) that fails in AR/SR — a faithful reproduction of the alpha-decay failure the paper's design targets. The V3 protocol implements the response: OOS-robust, parsimony-penalized selection; orthogonalization; portfolio-aware scoring; prompt/parser tightening; and plateau-aware compute. And we closed the loop: OOS-robust + parsimony-penalized selection (Result 4) repairs the CSI300 IC degradation (**+0.0007** raw robust vs baseline), while per-date orthogonalization shows a smaller but positive CSI500 residual lift (**+0.0007**; raw robust is **+0.0017**). CSI300 orthogonalization is too aggressive in this run (**-0.0035**), so V3's lesson is universe-aware gating, not blind residualization. Result 6 then exercises the final protocol items on the same run: a portfolio-aware objective re-ranks toward lower-turnover, regime-stable candidates; the prompt/parser contract finds no hidden knobs in the refreshed best Kimi factor; and plateau-aware early-stopping would save 31% of compute at patience=30. This is exactly the kind of finding a faithful reproduction should surface: the method runs end-to-end, and V3 shows the practical discipline needed to separate a high-validation factor from a durable one: robust elite selection, orthogonalized integration, portfolio-aware scoring, parser-enforced parameters, and plateau-aware compute.

Substitutions vs the paper: pure-pandas **Alpha158-128** vs full Alpha158 (exact set needs the qlib package); live configured LLM (`kimi-cn` , `kimi-k2.6` , `https://api.moonshot.cn/v1` , `n_llm=11`) vs paper Gemini-2.5-Pro; 300-iteration budget (between the paper's 200/400 settings, not the full 400-run grid); report-bootstrapping (FE-report) out of scope → FE-alpha variant. Data IS real (Qlib `.bin` bundle, 2008–2024). See REPRODUCTION_PLAN.md / REAL_DATA.md.